SHIFTSURE

Product Requirements Document

Fair Workforce Reallocation During Automation and Disruption

A build-ready specification for an almost-production-ready, fully working prototype.

DOCUMENT DETAILS	VALUE
Version	1.0
Status	Prototype specification
Domain	AI-assisted workforce decision support
Hackathon	SAP Hackfest Hackathon Bengaluru 2026
Theme	Inclusive Workforce
Data policy	Simulated workforce data; no live SAP integration claimed
Deployment	Not required for prototype acceptance

Product principle

When work disappears, ShiftSure helps organizations determine who can do the remaining work, what training is required, what the alternatives cost, and whether the proposed plan distributes disruption and opportunity fairly.

Implementation priority: real optimization and deterministic calculations first; end-to-end workflow second; AI explanations after the core system is reliable.

1. Executive Summary

ShiftSure is a workforce decision-support platform designed to help organizations redeploy employees when automation, AI adoption, or operational disruptions reduce the amount of available work. Instead of immediately sending workers home, reducing overtime opportunities, or recruiting externally, ShiftSure identifies available internal capacity, matches workers to suitable open tasks, evaluates short training bridges, and measures fairness and economic implications.

The platform combines deterministic business rules, mathematical optimization, workforce data, fairness analytics, and an explanation layer. Managers retain control over every final decision.

Prototype boundaries

- One simulated plant with two to three production lines and approximately 50–100 simulated workers.
- Configurable disruption scenarios and idle-hours calculation.
- Actual constraint-based workforce optimization, not random assignments.
- Internal redeployment, training recommendations, and internal-versus-external hiring comparisons.
- Fairness metrics, counterfactual testing, manager approval, worker-facing views, and audit history.
- Local execution; cloud deployment, live SAP integration, and production infrastructure are not mandatory.

2. Problem Statement

When automation or disruption reduces the workload on a production line, managers must decide whether to redeploy workers, provide training, reduce overtime, or recruit externally. Decisions made under time pressure and with incomplete information can leave internal capacity unused, distribute lost work unevenly, overlook training opportunities, or create safety and transparency risks.

Problems ShiftSure addresses

- **Underutilized capacity:** employees have available hours while other teams have unfilled work.
- **Unequal disruption:** idle hours, lost overtime, and training access may be distributed unevenly.
- **Missed internal opportunities:** potentially suitable employees may be overlooked in favor of external recruitment.
- **Safety and compliance risk:** assignments may violate certification, shift, availability, or location rules.
- **Insufficient transparency:** workers and managers may not know why an assignment was recommended or rejected.

3. Goals, Non-Goals, and Scope

3.1 Product goals

ID	Goal	Expected outcome
G-01	Reduce avoidable idle hours	Calculate idle hours before and after reallocation.
G-02	Increase task coverage	Match eligible workers to open tasks.
G-03	Prioritize internal talent	Identify training bridges before external hiring.
G-04	Enforce constraints	Prevent recommendations that violate hard rules.
G-05	Measure fairness	Compare idle time, overtime loss, and training access.
G-06	Explain decisions	Provide evidence-based reasons for matches and exclusions.
G-07	Preserve human control	Require authorized manager approval.
G-08	Ensure reproducibility	Persist inputs, plan versions, results, and audit history.
G-09	Demonstrate value	Compare three allocation strategies using consistent metrics.

3.2 Non-goals

- No real employee actions, automated transfers, dismissals, promotions, or external hiring.
- No claims that fairness metrics prove fairness or legal compliance.

- No live SAP integration, multi-site planning, cloud deployment, or CI/CD requirement.
- No LLM-generated assignments, fabricated metrics, or LLM overrides of hard constraints.
- No inference of sensitive employee attributes from surveillance or behavioral data.

4. Target Users

Persona	Needs	Primary actions
Plant / Operations Manager	Restore coverage and compare plans.	Create scenarios, review assignments, approve/reject plans.
HR / Workforce Analyst	Evaluate internal talent, training, cost, and disparities.	Review skills, hiring comparisons, fairness metrics, and what-if tests.
Employee / Worker	Understand proposed assignments and correct records.	View authorized information, read explanations, submit corrections.
Demo Operator / Admin	Maintain a repeatable simulated environment.	Load/reset data, configure scenarios, inspect run status and errors.

5. Feature Priorities

Feature	Priority	Acceptance scope
Simulated plant and workforce dataset	P0	Seed 50–100 workers and tasks.
Disruption and idle-hours calculation	P0	Scenario inputs produce tested calculations.
Constraint-based optimizer	P0	Actual solver, hard-rule validation, explicit infeasibility.
Three-way plan comparison	P0	Baseline, optimizer without fairness, fairness-balanced plan.
Training and hiring comparison	P0	Explicit costs, lead times, assumptions.
Fairness dashboard and counterfactual	P0	Computed metrics, group denominators, controlled what-if.
Manager approval and audit log	P0	Persisted state transitions and history.
Worker view and correction requests	P0	Authorized view and auditable correction workflow.
Evidence-grounded LLM explanations	P1	Add after deterministic core; fallback required.
CSV import/export	P1	Implement where practical.
Live SAP, cloud deployment, multi-site planning	Out of scope / future	Not required for prototype acceptance.

6. Functional Requirements

6.1 Workforce and task data

- **FR-001:** Worker records include unique ID, role, line/site, shift, available hours, skills, proficiency, certification status, employment category, and relevant overtime data.
- **FR-002:** Task records include ID, destination, demand/headcount, required skills, mandatory certifications, shift, hours, and training/cost assumptions.
- **FR-003:** Reject or flag duplicate IDs, negative hours, invalid dates, missing mandatory fields, and impossible requirements.
- **FR-004:** Clearly label seeded data and generated results as simulated in the UI and exports.
- **FR-005:** Workers can submit skill corrections; changes remain pending until authorized review. Approved changes are auditable and can trigger re-planning.

6.2 Disruption scenarios

- **FR-006:** Managers can create or configure an automation or operational disruption.
- **FR-007:** Scenario parameters include affected lines, workload reduction or remaining workload, planning period, and optional duration.
- **FR-008:** Calculate idle capacity using workload, existing commitments, and planning-period assumptions; do not treat an entire shift as idle automatically.

- **FR-009:** Scenario execution must not silently alter base workforce data or existing approved plans.

6.3 Reallocation optimizer

- **FR-010:** Use Google OR-Tools or an equivalent real constraint solver against actual records.
- **FR-011:** Enforce certification, skill eligibility, shift, availability, location, capacity, non-overlap, and task-demand rules.
- **FR-012:** Support configurable objectives for coverage, idle hours, training burden, external hiring, and fairness.
- **FR-013:** Persist solver status, assignments, unfilled demand, objective settings, and relevant inputs.
- **FR-014:** Explicitly report infeasible, timed-out, feasible, or optimal outcomes; never fabricate assignments.
- **FR-015:** Use versioned inputs and document solver behavior where multiple equivalent solutions exist.

6.4 Three-way comparison

- **FR-016:** Compare (1) a documented deterministic supervisor-style baseline, (2) optimization without fairness balancing, and (3) fairness-balanced optimization.
- **FR-017:** Evaluate all plans using consistent definitions: idle hours, task coverage, internal fill, unfilled demand, training cost/time, hiring cost/lead time, overtime loss where available, and fairness metrics.
- **FR-018:** Explain trade-offs; do not claim one plan is universally best regardless of priorities.

6.5 Training and hiring

- **FR-019:** Identify workers who qualify now or may qualify after a permitted training bridge.
- **FR-020:** Show missing skills, training duration, estimated cost, and remaining qualification requirements.
- **FR-021:** Compare internal training with configurable external recruitment cost and lead-time estimates.
- **FR-022:** Present options for human review; do not initiate hiring or promise qualification.

6.6 Fairness and counterfactual testing

- **FR-023:** Calculate idle hours, overtime loss where data exists, training access, and relevant redeployment opportunities.
- **FR-024:** Compare configured groups such as contract/permanent and shifts; use demographic data only when appropriate and available.
- **FR-025:** Make review thresholds configurable and identify them as policy settings.
- **FR-026:** Flag disparities for review, not as proof of discrimination.
- **FR-027:** Fairness objectives cannot override hard safety, certification, or eligibility constraints.
- **FR-028:** Run a counterfactual changing one non-job attribute while holding job-relevant factors constant.
- **FR-029:** Show assignment changes, affected workers/tasks, metric differences, and changed/unchanged assumptions.
- **FR-030:** Use equivalent solver settings and account for tie-breaking among equally optimal solutions.
- **FR-031:** Treat changes as review signals, not conclusive proof of unfairness; unchanged outcomes do not prove fairness.

6.7 Explanations, approvals, and audit

- **FR-032:** Explain recommendations from recorded attributes, task requirements, rules, and solver evidence.
- **FR-033:** LLMs may verbalize structured evidence but may not calculate authoritative metrics, invent facts, change results, override rules, or approve plans.
- **FR-034:** Provide deterministic template-based explanations if the LLM is unavailable or returns invalid output.
- **FR-035:** Each explanation should reference the supporting assignment, rule, constraint, or calculation.
- **FR-036:** Managers can review assignments, unfilled tasks, training, costs, fairness flags, and solver status.
- **FR-037:** Support Draft, Generated, Pending Approval, Approved, Rejected, and Superseded states with valid transitions.
- **FR-038:** Only authorized managers can approve/reject; rejection requires a recorded reason.
- **FR-039:** Submitted plans preserve reviewed inputs/results; material changes require a new version or renewed review.
- **FR-040:** Audit scenario changes, solver runs, correction requests, rule changes, and approval decisions.

- **FR-041:** Approval changes prototype status only; it does not execute real-world transfers or update SAP.

7. UX and Interface Requirements

Manager workspace

- Overview: scenario, affected lines, available hours, open tasks, and planning status.
- Scenario configuration: create and edit disruption events.
- Plan comparison: side-by-side strategy comparison.
- Assignment review: matches, eligibility evidence, training, and unfilled tasks.
- Fairness analysis: metrics, group sizes, thresholds, and review flags.
- Hiring comparison: training and recruitment costs and time estimates.
- Approval and history: state, approval controls, versions, and audit events.

Worker workspace

- Show only the worker's authorized proposed/approved assignment information.
- Explain recommendation or non-selection where evidence supports it.
- Show recorded skills/certifications and a correction request form with status.
- Clearly distinguish proposed assignments from manager-approved decisions.

Interaction quality

- Implement loading, empty, success, warning, and error states.
- Every visible action has a defined outcome; forms validate inputs.
- Use meaningful units and labels; never rely on color alone.
- Explain fairness flags and confirm destructive actions.

8. System Architecture

Layer	Technology	Responsibility
Frontend	React + TypeScript	Manager/worker interfaces and API integration.
Backend	Python + FastAPI	Validation, APIs, workflow, permissions.
Optimization	Google OR-Tools	Constraint-based allocation.
Database	SQLite	Persistent prototype records and history.
Analytics	Python	Idle hours, hiring comparisons, fairness metrics.
Explanation	Optional LLM + deterministic fallback	Evidence-grounded natural language only.
Testing	Pytest + frontend test tools	Unit, integration, regression, and end-to-end tests.

Logical services

- Data/validation service: workers, tasks, skills, certifications, shifts.
- Scenario engine: disruption configuration and workload impact.
- Optimization service: solver model, validation, status, and results.
- Hiring comparison service: internal training and external estimates.
- Fairness analytics: metrics, group comparisons, thresholds, counterfactuals.
- Explanation service: natural language over structured evidence.
- Approval/audit service: permissions, state transitions, versions, history.
- Frontend: accessible views and API-driven workflows.

Architecture principles: keep business logic out of UI components; use one authoritative source per metric; validate solver output independently; persist plan snapshots; make failure states explicit; run locally without cloud services.

9. Data Model

Entity	Key fields / purpose
Worker	ID, display name, role, site, shift, employment type, availability.
Skill / WorkerSkill	Skill definitions, proficiency, verification status.
Certification	Worker, certification type, validity dates, status.
Task	Role, site, shift, skills/certifications, hours, headcount.
DisruptionScenario	Event type, affected lines, workload change, planning period.
TrainingBridge	Worker/task, missing skills, duration, estimated cost.
OptimizationRun	Scenario, solver status, objective settings, timestamps.
Plan / Assignment	Strategy, version, state; worker-task hours and eligibility evidence.
MetricSnapshot	Plan, metric name/value, group, denominator, definition.
CounterfactualRun	Baseline run, changed attribute, comparison results.
CorrectionRequest	Worker, requested correction, status, reviewer.
Approval	Plan, manager, decision, reason, timestamp.
AuditEvent	Actor, action, entity, timestamp, event details.
HiringEstimate	Role/task, recruitment cost, lead time, assumptions.

Integrity rules: unique IDs; validated foreign keys; no over-capacity or overlapping assignments; valid plan references; retained superseded versions; metrics tied to plan definitions; and restricted access to sensitive attributes.

10. Optimization Model

Decision variables

Let x■■ represent the hours assigned to worker i for task j. Binary assignment variables may be added for tasks requiring discrete worker selection.

Hard constraints

Worker capacity: Σ■ x■■ ≤ H■, where H■ is available capacity in the planning period.

Task demand: Σ■ x■■ ≤ D■, where D■ is task demand. Minimum required coverage can be modeled separately.

- Ineligible worker-task combinations receive no assignment.
- Mandatory certifications must be valid for the task.
- Shift, availability, location, capacity, and non-overlap rules must be enforced.
- Training only changes eligibility where task rules explicitly permit qualification after training.

Objective strategy

A conceptual configurable objective is:

Maximize: w■C – w■I – w■T – w■H – w■F

C = covered demand; I = avoidable idle hours; T = training burden; H = external hiring burden; F = configured fairness penalty; w values = documented weights. This is a conceptual model, not a mandate to combine every metric into an uncalibrated score. Avoid double-counting equivalent costs. Safety and mandatory eligibility remain hard constraints.

Solver validation

- Validate certifications, capacity, shifts, overlaps, task eligibility, demand, and training pathways independently.
- Reject plans that fail validation.

- Report optimal, feasible, infeasible, or timed-out states accurately.

11. Fairness Measurement

Metric	Method	Interpretation safeguard
Idle hours	Group total and average per eligible worker.	Compare denominators and absolute/relative gaps.
Overtime loss	Baseline opportunities versus scenario outcome, when data exists.	Missing data is unavailable, not zero.
Training access	Eligible workers offered training and recommendation rates.	Distinguish offers from successful qualification.
Counterfactual sensitivity	Compare assignments after controlled attribute change.	Account for tie-breaking and equivalent solutions.

Show group sizes, denominators, missing-data warnings, and definitions. Thresholds are review policies, not universal fairness standards. Disparities warrant investigation but do not by themselves prove discrimination or legal non-compliance.

12. API Contract

Method	Endpoint	Purpose
GET	/api/health	Local application health.
GET / POST	/api/workers	List/create workers.
GET	/api/workers/{id}	Retrieve authorized worker record.
POST	/api/corrections	Submit skill correction.
GET	/api/tasks	List open tasks.
POST	/api/scenarios	Create disruption scenario.
GET	/api/scenarios/{id}	Read scenario.
POST	/api/scenarios/{id}/calculate	Calculate workload impact.
POST	/api/scenarios/{id}/optimize	Generate optimization run.
GET	/api/runs/{id}	Read run status/results.
GET	/api/runs/{id}/compare	Three-way comparison.
GET	/api/plans/{id}	Plan and assignment details.
GET	/api/plans/{id}/fairness	Fairness metrics.
POST	/api/plans/{id}/submit	Submit for approval.
POST	/api/plans/{id}/approve	Authorized approval.
POST	/api/plans/{id}/reject	Reject with reason.
POST	/api/counterfactuals	Run what-if comparison.
GET	/api/audit	Authorized audit history.
GET	/api/reports/{id}	Retrieve report.

API behavior: validate inputs; use appropriate status codes; enforce backend authorization; reject invalid state transitions; distinguish business-rule failures from server errors; protect private worker data; expose solver failures explicitly. Long-running runs may use status polling if synchronous execution is impractical.

13. Security, Privacy, and Reliability

Security

- Enforce role-based authorization in the backend; never rely on frontend-only checks.
- Validate and sanitize input; use safe database access patterns.
- Keep secrets outside source control and LLM credentials outside application code.
- Restrict sensitive worker data and audit history.

Privacy

- Use fictional identities and simulated attributes.
- Collect only data needed for the prototype.
- Do not infer sensitive attributes or send unnecessary personal data to an LLM.
- Separate sensitive attributes from operational assignment inputs wherever practical.

Reliability

- Handle missing/inconsistent data explicitly and roll back incomplete transactions.
- Preserve plan versions and prevent misleading approval states after errors.
- Use deterministic seed/reset procedures.
- Log failures without unnecessarily logging sensitive data.

The prototype is not certified for real employment decisions. Real-world use would require legal, privacy, security, operational, and workforce-policy review.

14. Non-Functional Requirements

ID	Requirement	Acceptance expectation
NFR-001	Usability	Core manager workflow without developer intervention.
NFR-002	Integrity	Reject invalid assignments and state transitions.
NFR-003	Auditability	Trace material changes to events.
NFR-004	Reproducibility	Equivalent versioned inputs yield equivalent explainable outcomes.
NFR-005	Maintainability	Separate optimizer, metrics, APIs, and UI.
NFR-006	Performance	50–100-worker seeded scenario completes in practical local-demo time.
NFR-007	Error handling	Actionable errors, no silent failures.
NFR-008	Accessibility	Keyboard use, clear labels, status not conveyed by color alone.
NFR-009	Data isolation	Workers cannot access unauthorized records.
NFR-010	Calculation correctness	Documented formulas with independent tests.

15. Evaluation and Success Metrics

Operational metrics

Idle hours avoided = baseline idle hours – plan idle hours.

Task coverage = covered demand ÷ total required demand × 100. Define demand consistently in hours, units, or task slots.

Internal fill rate measures the share of covered demand or filled positions satisfied by existing employees. State the denominator. Compare training and hiring estimates using explicit costs, lead times, and assumptions.

Fairness and quality

- Maximum group-level difference in average idle hours.
- Overtime-loss differences when baseline data exists.
- Training-offer-rate differences among eligible workers.
- Assignment sensitivity in counterfactual testing.
- Hard-constraint violations in accepted plans: zero.
- Recommendations with traceable explanations.
- Correct rejection of invalid inputs and unauthorized approval attempts.
- Successful completion of core end-to-end tests.

Repeatable demo scenario

- Simulate an automation rollout that substantially reduces workload on one line.
- Calculate idle capacity and identify open tasks.
- Generate and compare the three allocation strategies.
- Show operational metrics, fairness outcomes, and trade-offs.
- Identify training bridges and compare with external hiring estimates.
- Run a counterfactual test.
- Submit the plan for manager review and approval/rejection.
- Show worker-facing explanations and audit history.

Every result must be generated by the implemented system and labelled as simulated. No impact figures should be claimed before measurement.

16. Testing Strategy

Test level	Required coverage
Unit	Idle-hours math, task demand, certification validity, capacity, hiring estimates, fairness formulas, counterfactuals, state transitions.
Optimizer	Feasible allocation, missing certification, no eligible candidate, insufficient capacity, shift/location restrictions, training bridge, fairness trade-off, equivalent optima, timeout/infeasibility.
API/integration	Scenario-to-plan flow, persistence, role authorization, worker data isolation, correction review, approvals, audit, error recovery.
Frontend	Scenario validation, comparison, assignment review, fairness states, worker corrections, approvals, loading/error/empty states.
End-to-end	Complete workflow from disruption creation to manager decision with metrics, persistence, explanations, and audit history.

17. Acceptance Checklist

- ■ Application starts locally with documented setup instructions.
- ■ Seed data creates two to three lines and approximately 50–100 simulated workers.
- ■ Disruption scenarios and tested idle-hours calculations work.
- ■ Real optimizer generates assignments and respects hard constraints.
- ■ Infeasible scenarios are reported accurately; no fabricated assignments.
- ■ Three comparable plans are generated and evaluated.
- ■ Training and external hiring are compared with explicit assumptions.
- ■ Fairness metrics and controlled counterfactual tests work.
- ■ Managers can approve/reject through the interface with authorization.
- ■ Workers can view authorized data and submit skill corrections.
- ■ Plan versions and material actions persist in the audit log.
- ■ Explanations use real evidence; the core workflow works without an LLM.
- ■ Automated tests cover critical logic, constraints, permissions, and state transitions.
- ■ Simulated data and estimates are clearly labelled.
- ■ No real-world workforce actions or live SAP integration are claimed.

18. Implementation Roadmap

Phase	Work	Exit condition
1. Foundation	Repository, local environment, schema, seed data, validation, APIs, navigation.	Simulated workforce and tasks load and validate.
2. Core logic	Disruption scenarios, idle hours, hard rules, OR-Tools model, solver validation.	Scenario yields a valid plan or explicit infeasibility.
3. Decision intelligence	Three strategies, hiring comparison, fairness metrics, counterfactual.	Three plans compared using tested calculations.
4. Product workflow	Manager/worker views, approvals, corrections, versioning, audit, authorization.	Main workflow runs through the UI without manual DB edits.
5. Quality/demo	LLM fallback, automated tests, edge cases, UI states, repeatable demo.	Reliable end-to-end demonstration with no fabricated outputs.

19. Future Scope

- Authorized SAP SuccessFactors and scheduling integrations.
- Approved import of real workforce data.
- Multi-site redeployment and larger solver models.
- Customer-configurable labor rules and policy packs.
- Advanced fairness objectives and sensitivity analysis.
- Enterprise identity, audit, backup, monitoring, and deployment infrastructure.

These are future possibilities, not prerequisites for prototype acceptance.

20. Product Differentiation

- Reallocation after work disappears, not only filling staffing shortages.
- Measurable fairness and explicit trade-offs rather than unverified AI claims.
- Training and internal hiring comparisons before external recruitment.
- Worker-facing explanations and correctable records.
- Counterfactual sensitivity testing for non-job attributes.
- Human approval and traceable decisions.
- Separation of solver outputs from LLM explanations.

The claim that these capabilities are unique relative to SAP or partner products must be validated before being presented as established market fact.

21. Final Definition of Done

ShiftSure is complete as a prototype when it can take a simulated automation disruption, calculate workforce impact, produce and validate alternative reallocation plans, compare internal training with external hiring, measure fairness, explain recommendations, support worker corrections, and record an authorized manager's decision.

> **A locally runnable, fully functional decision-support application with real optimization, deterministic metrics, human approval, persistent history, test coverage, and a repeatable demonstration using clearly labelled simulated data.**

Deployment and live enterprise integration are deliberately excluded from the definition of done.